

E-commerce Warehouse Location Optimization for E-Commerce in India

Facility Location Problem

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Motivation...

- Traditional heuristics approaches to warehouse location often result in **suboptimal facility networks** that increase operational cost.
- **Fails** to account for the inherently **stochastic nature of customer demand**.
- E-Commerce in India requires strategic facility placement to serve diverse urban and rural markets efficiently.

Research Question

How can we minimize the cost of serving stochastic customer demand by optimally choosing warehouse locations from a finite set of candidate sites?



Modeling it as an **uncapacitated facility location problem (FLP)**, using **mixed integer linear programming (MILP)** with stochastic elements to determine the optimal subset of warehouses to open.

- **What to measure ?**

- Coordinates of the demand point (x_i, y_i) and candidate warehouse (x_j, y_j) .
- Expected demand at demand points (λ_i) .
- Facility to demand point unit transportation cost (c_{ij}) .

- **How to measure ?**

We assume:

- $\lambda_i \overset{ind}{\sim} \text{Uniform}(a, b)$ (demand follows some distribution)
- $c_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$ (Euclidean distances)



- **Study design :**

Simulating a metropolitan region $[0, 100]^2$ units containing 50 customer demand points (with expected demand λ_i at i) and 7 candidate warehouse locations (with identical fixed setup cost of 1,000 units each).

We assume : $x_i, y_i, x_j, y_j \stackrel{ind}{\sim} Uni(0, 100)$ and $\lambda_i \stackrel{ind}{\sim} Uni(50, 200)$.

I : Set of customer locations, indexed by i and J : Set of potential warehouse locations, indexed by j .

Demand at a point follows a Poisson

Actually, if the demand at i is $d_i \stackrel{ind}{\sim} Pois(\lambda_i)$, then the mean is λ_i . We then simulate λ_i from $ind. Uniform(a, b)$.



Analysis : Optimization Formulation

Parameters

- λ_i, c_{ij} (Expected demand at i , Transportation cost of (i, j))
- F_j : Fixed cost of opening warehouse j
- M_j : A large constant, representing warehouse j 's capacity (effectively uncapacitated)

Decision Variables

- $y_j \in \{0, 1\}$: 1 if warehouse j is opened, 0 otherwise
- $x_{ij} \geq 0$: Units of demand at customer i satisfied by warehouse j

Objective Function: (Minimize total cost)

$$\min \sum_{j \in J} F_j y_j + \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij}$$

Subject to constraints:

- Demand satisfaction:

$$\sum_{j \in J} x_{ij} = \lambda_i, \quad \forall i \in I$$



Analysis : Results and Visualizations

- Capacity (Big-M) constraints: (effectively uncapacitated)

$$\sum_{i \in I} x_{ij} \leq M_j y_j, \quad \forall j \in J$$

- Strong linking constraints:

$$x_{ij} \leq \lambda_i y_j, \quad \forall i \in I, j \in J$$

- Variable domains:

$$y_j \in \{0, 1\}, \quad x_{ij} \geq 0$$

Results: This model (MILP model) which is **not a convex optimization model** was implemented and solved using the `ompr` and `ROI.plugin.symphony` packages in R and the full implementation of this model can be accessed at: [Google Colab Notebook](#). For a particular generated setup **with 5 warehouse candidates**, these are the results:

- Optimal cost: 4,753.28 (in scaled units)
- **Warehouses opened: 2 (3 closed)**
- Naive approach (open all warehouses): 5,730.15 (in scaled units)
- **Cost savings from optimization: 17.047%**



Analysis : Visualizations

.... with 7 warehouse candidates, these are the results:

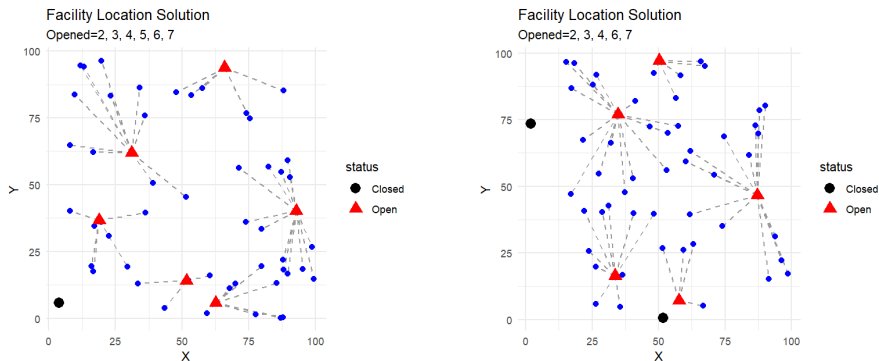


Figure: Optimal warehouse locations (red triangles) and customer assignments (blue dots). Dashed lines indicate assignment of each customer to its serving facility.



Conclusion

Insights:

- **Selective Facility Opening**
- **Stochastic Consideration**
- **Cost Efficiency**
- **Demand-Driven Assignments**

Limitations:

- **Distance Metric**
- **Homogeneous Facilities**
- **Demand Distribution**
- **Single-Period Model**



- **Network-Based Distances**
- **Multi-Period Modeling**
- **Empirical Demand Patterns**
- **Service Level Constraints**
- **Multi-Echelon Extensions**

In conclusion, this study demonstrates the value of mathematical optimization for strategic logistics decisions in e-Commerce.

References

- 1 Optimal Facility Location – Wikipedia.
- 2 SCIP Optimization Suite Documentation.
- 3 Convex Optimization Stephen Boyd, Lieven Vandenberghe



Thank You

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